Software Engineering Lab - 01

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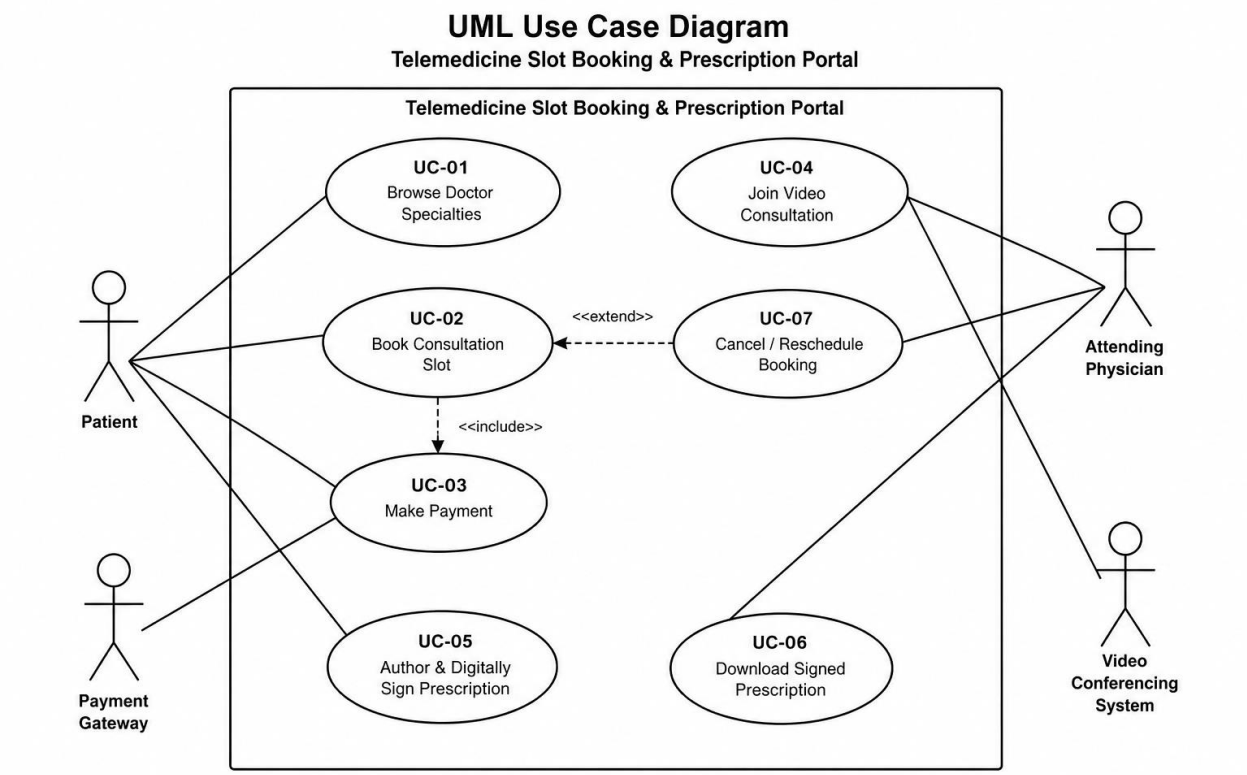
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Section: G

Problem Statement: Telemedicine Slot Booking & Prescription Portal

Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records.	High	Pass: Valid cryptographic signature attached to prescription PDF. Fail: Prescription generated without doctor license validation.	Ensures legal validity, authenticity, and traceability of prescriptions issued remotely.	Given (instructor-provided)
FR-002	Functional	The system shall allow patients to search and filter available physicians by specialty, language, and next available slot.	High	Pass: When a patient applies a specialty filter (e.g., "Cardiology"), only matching physicians are displayed within 2 seconds. Fail: Unfiltered or irrelevant physicians remain listed.	Reduces patient effort in locating a suitable physician and speeds up access to care.	
FR-003	Functional	The system shall allow a patient to book an available video consultation slot for a selected physician on a chosen date and time.	High	Pass: Upon confirmation, the slot status changes to "Booked" and is immediately removed from availability for all other patients. Fail: Slot remains bookable by more than one patient (double-booking).	Core scheduling function; prevents double-booking and guarantees the physician's calendar integrity.	
FR-004	Functional	The system shall generate and deliver an encrypted, time-limited tele-consultation room link to both patient and physician at least 10 minutes before the scheduled session.	High	Pass: Both parties receive a working link, valid only within a ±15-minute window around the slot time, at least 10 minutes before start. Fail: Link is delivered late, is reusable outside the window, or is accessible without authentication.	Provides secure, timely access to the consultation and prevents unauthorized access to the video session.	
FR-005	Functional	The system shall allow a patient to view and download their digitally signed prescription PDF from their account after the consultation concludes.	Medium	Pass: Within 5 seconds of the physician signing, the patient can download an identical PDF (verified by matching cryptographic hash). Fail: Downloaded PDF differs from the signed original or is unavailable after 5 seconds.	Gives patients timely, verifiable access to their own treatment and prescription records.	
NFR-001	Nonfunctional (Performance & Security)	All video session metadata and patient clinical notes must be encrypted in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Benchmarking tests confirm TLS 1.3 in transit and AES-256 at rest, meeting target latency under simulated peak load. Fail: Any data found unencrypted in transit or at rest, or latency exceeds target under load.	Protects sensitive health data and ensures regulatory compliance (e.g., HIPAA/GDPR).	Given (instructor-provided)
NFR-002	Nonfunctional (Scalability & Availability)	The system shall support at least 500 concurrent video consultation sessions while maintaining 99.5% uptime during business hours (8 AM–8 PM).	High	Pass: Load testing with 500 simulated concurrent sessions shows zero dropped sessions, and 30-day uptime logs show ≥ 99.5% availability during business hours. Fail: Sessions drop under 500-user load, or uptime falls below 99.5%.	Ensures the platform remains reliable and scalable during peak patient demand periods.	



Use-Case Flow Specification:

Use case ID	**UC - 02**
Use Case Name	Book Consultation Slot
Actors	Patient (primary); Payment Gateway (secondary, via included use case)
Triggers	The patient chooses to schedule a video consultation with a physician.

Preconditions

- The patient is registered and successfully logged in to the portal.
- The patient has completed any required identity/insurance verification.
- At least one physician has published open consultation slots in the system.

Postconditions

Success:

- The selected slot is reserved exclusively for the patient, payment is confirmed, a unique booking ID is created, and the physician's calendar is updated.

Failure:

- No slot is reserved, no payment is captured, and system state (slot availability) is unchanged from before the use case began.

Main Success Scenario

1. Patient selects "Book Consultation" and chooses a physician from the specialty list (see UC-01: Browse Doctor Specialties).
2. System displays the physician's available time slots for the next 14 days.
3. Patient selects a preferred date and time slot.
4. System re-validates that the selected slot is still available.
5. System displays the consultation fee and prompts the patient to choose a payment method.
6. The patient selects a payment method and confirms the booking.
7. System invokes the «include» use case "Make Payment," connecting to the Payment Gateway to process the transaction.
8. Payment Gateway authorizes the payment and returns a success confirmation.
9. System marks the slot as "Booked," generates a unique booking ID, and updates the physician calendar.
10. System displays a booking confirmation to the patient and notifies both patient and physician by email/SMS.
11. Use case ends successfully.